# StudyMate Test Document: Neural Networks and Machine Learning

## Purpose

This document is a small sample for testing a Retrieval-Augmented Generation (RAG) pipeline. It contains facts and explanations that can be used to check PDF extraction, text chunking, retrieval, and question answering.

## 1. What Is a Neural Network?

A neural network is a machine learning model made of connected units often called neurons. These units are organized into layers: an input layer, one or more hidden layers, and an output layer. Each connection has a weight, and neurons may also use a bias. The network combines inputs using weights and biases, then applies an activation function to produce an output.

## 2. What Is Backpropagation?

Backpropagation is an algorithm used to calculate how much each weight in a neural network contributes to prediction error. It works by propagating error-related gradients backward from the output layer toward earlier layers. These gradients help determine how the weights should be adjusted during training.

## 3. What Is Gradient Descent?

Gradient descent is an optimization method that updates model parameters in a direction intended to reduce the loss function. The learning rate controls the size of each update. If the learning rate is too large, training may become unstable; if it is too small, learning may take a long time.

## 4. What Is an Embedding?

An embedding is a numerical vector representing information such as a word, sentence, or document passage. Embedding models are trained to place related meanings near one another in vector space. Similarity search can compare a question vector with stored passage vectors to retrieve relevant content.

## 5. What Is Retrieval-Augmented Generation (RAG)?

Retrieval-Augmented Generation combines information retrieval with a language model. First, the system searches a knowledge source for passages relevant to a user's question. Then it provides those passages to a language model as context so the model can generate an answer grounded in the retrieved material. Retrieval quality depends on document parsing, chunking, embeddings, and search configuration.

## 6. What Is ChromaDB?

ChromaDB is a vector database that can store document text, embeddings, and metadata, and retrieve relevant items using vector similarity search. An application must configure or supply an embedding function so documents and questions can be represented in a compatible vector space. ChromaDB is not itself the language model that writes the final answer.

## 7. What Is Metadata?

Metadata is additional information stored alongside a document chunk. Examples include the source filename, page number, topic, or date. In a RAG application, metadata can help identify where a retrieved passage came from and can support filtering search results.

## 8. Test Questions

Use these questions to test StudyMate:

1. What does backpropagation calculate?

2. What does the learning rate control in gradient descent?

3. How does RAG use retrieved passages?

4. What role does ChromaDB play in a RAG pipeline?

5. According to this document, what is the capital of Japan?

The final question is intentionally unanswerable from this document. A well-grounded RAG assistant should say that the document does not provide that information.